

PCAP Malware Investigation Report

Jobayer Bhuiyan
Network Forensics • Week 8
Instructor: Christopher Lewis

Evidence-based revision with packet, alert, and artifact screenshots

Executive Summary. The packet capture shows a likely malware infection chain in which the file start.exe was downloaded from external IP 173.171.132.82 to internal host 10.100.9.107. NetworkMiner displays the MZ header and PE structure expected of a Windows executable, VirusTotal shows the file was flagged as malicious by 63/72 engines, Wireshark shows follow-on HTTP POST traffic, and Security Onion generated a high-confidence malware command-and-control alert. Together, these artifacts support the conclusion that the endpoint was compromised and then communicated back to an external system.

Key Answers to the Assignment Questions

Question	Answer
What file was downloaded and what does VirusTotal say?	The downloaded file is start.exe. VirusTotal flagged the sample as malicious with a detection ratio of 63/72 engines in the screenshot captured during analysis.
When did this happen and what machines were involved?	The Security Onion alert shows activity at 2018-10-31 15:36:17 UTC. The main systems involved were internal host 10.100.9.107 and external IP 173.171.132.82.
Who is the user that downloaded the file?	A specific human username is not visible in the provided screenshots. Based on the packet and alert evidence, the activity can be tied to endpoint 10.100.9.107, but additional host or authentication logs would be needed to identify the exact user with confidence.
What is the secondary infection and what is it generally there to do?	The observed follow-on traffic is consistent with TrickBot-style post-infection behavior. TrickBot is a modular trojan that can perform reconnaissance, steal credentials, communicate with command-and-control infrastructure, and download additional malware or follow-on payloads.

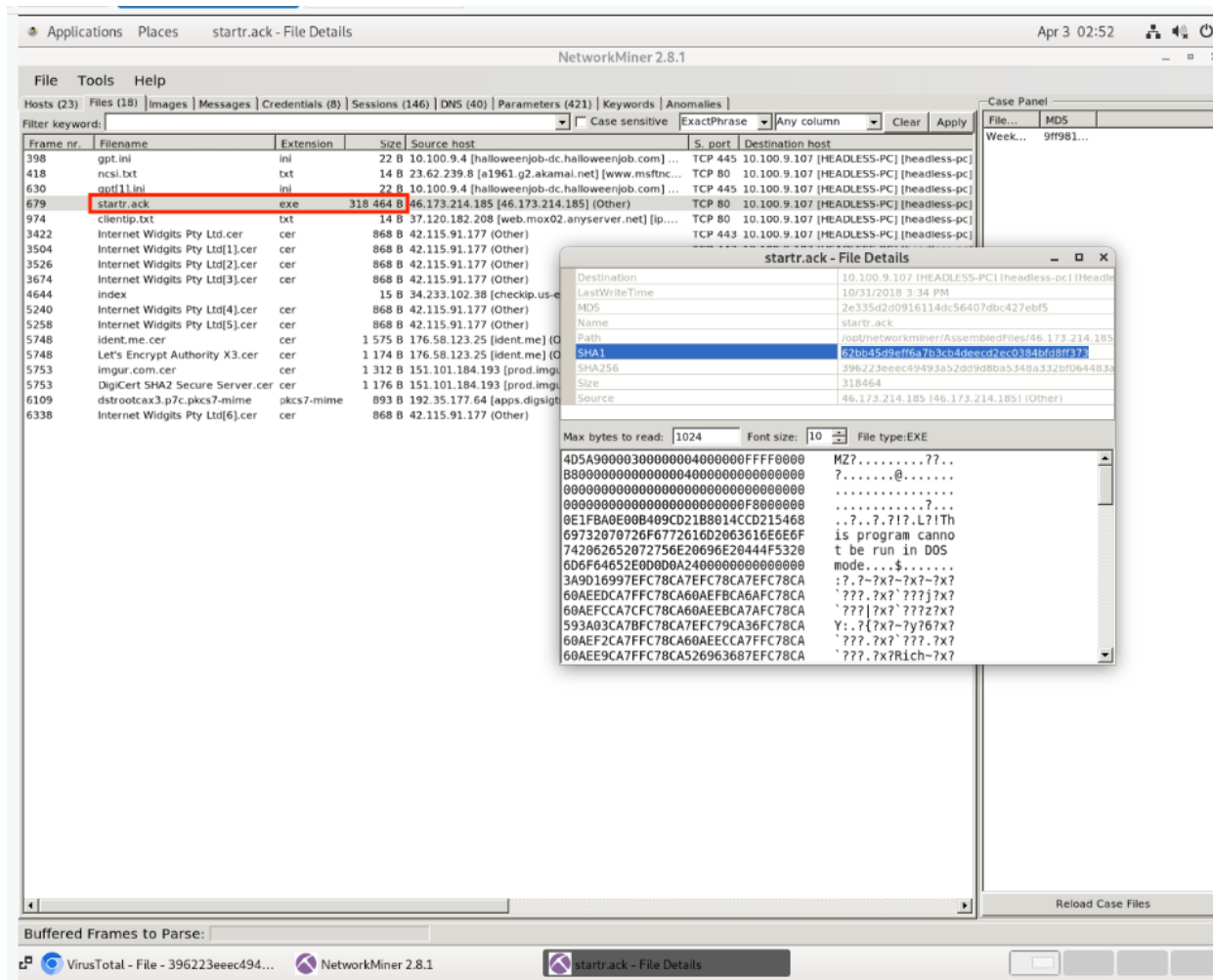
Methodology

The PCAP was examined in Wireshark to review HTTP conversations, exported objects, and POST traffic. NetworkMiner was used to recover transferred artifacts and inspect file content. Security Onion alerts were reviewed to correlate timestamped detection data with the network evidence. The suspicious file and the related external IP were then checked in VirusTotal to strengthen the reputation-based assessment.

Findings

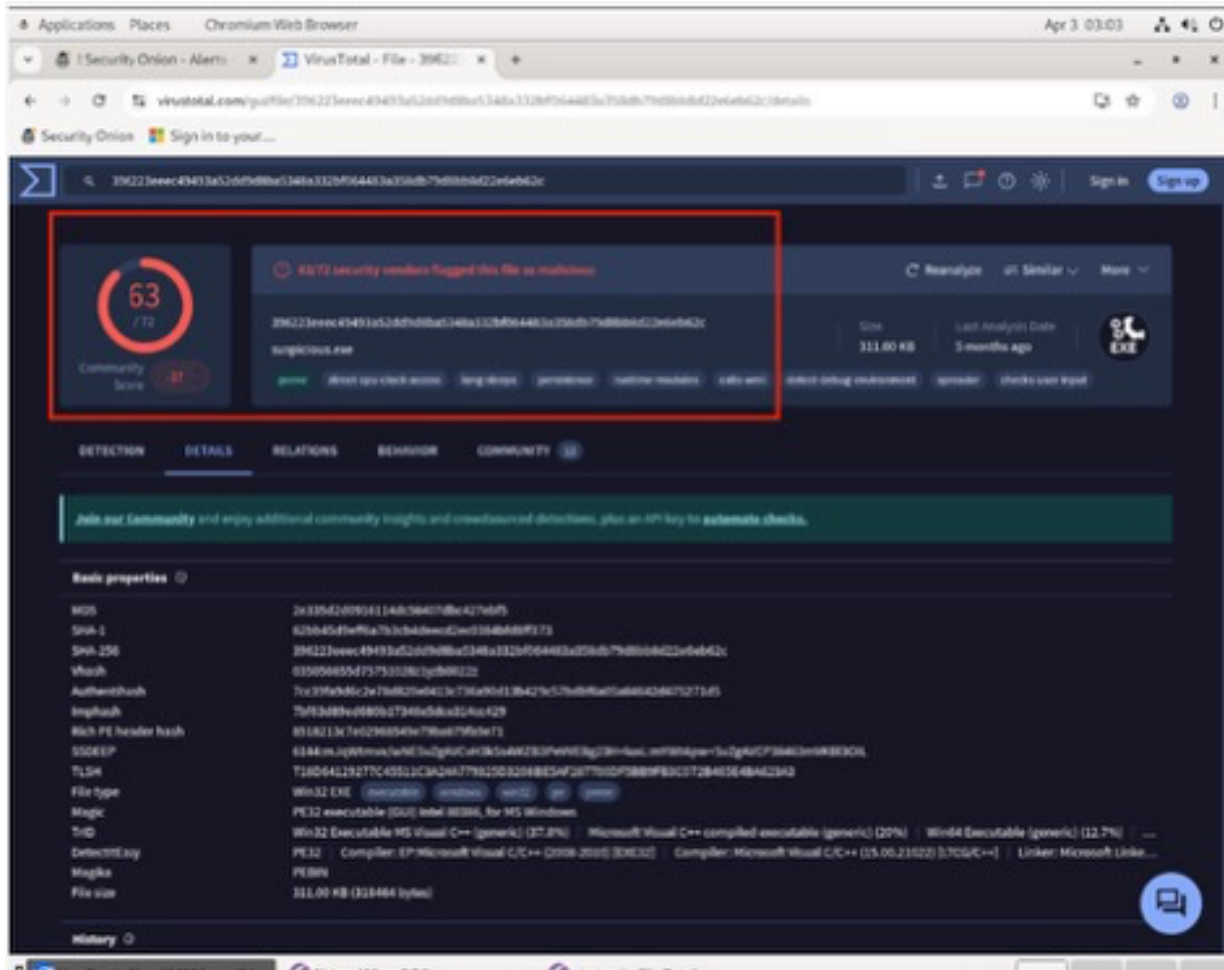
The evidence supports a multi-stage compromise narrative: a Windows executable was transferred to an internal endpoint, the file was independently flagged as malicious by VirusTotal, and the same endpoint subsequently generated HTTP POST traffic and a Security Onion alert categorized as malware command-and-control activity. Because the screenshots do not expose every payload byte or a named local user account, the conclusions below are written carefully and tied only to what the evidence directly supports.

Figure 1. NetworkMiner artifact view showing start.exe and the MZ/PE file signature.



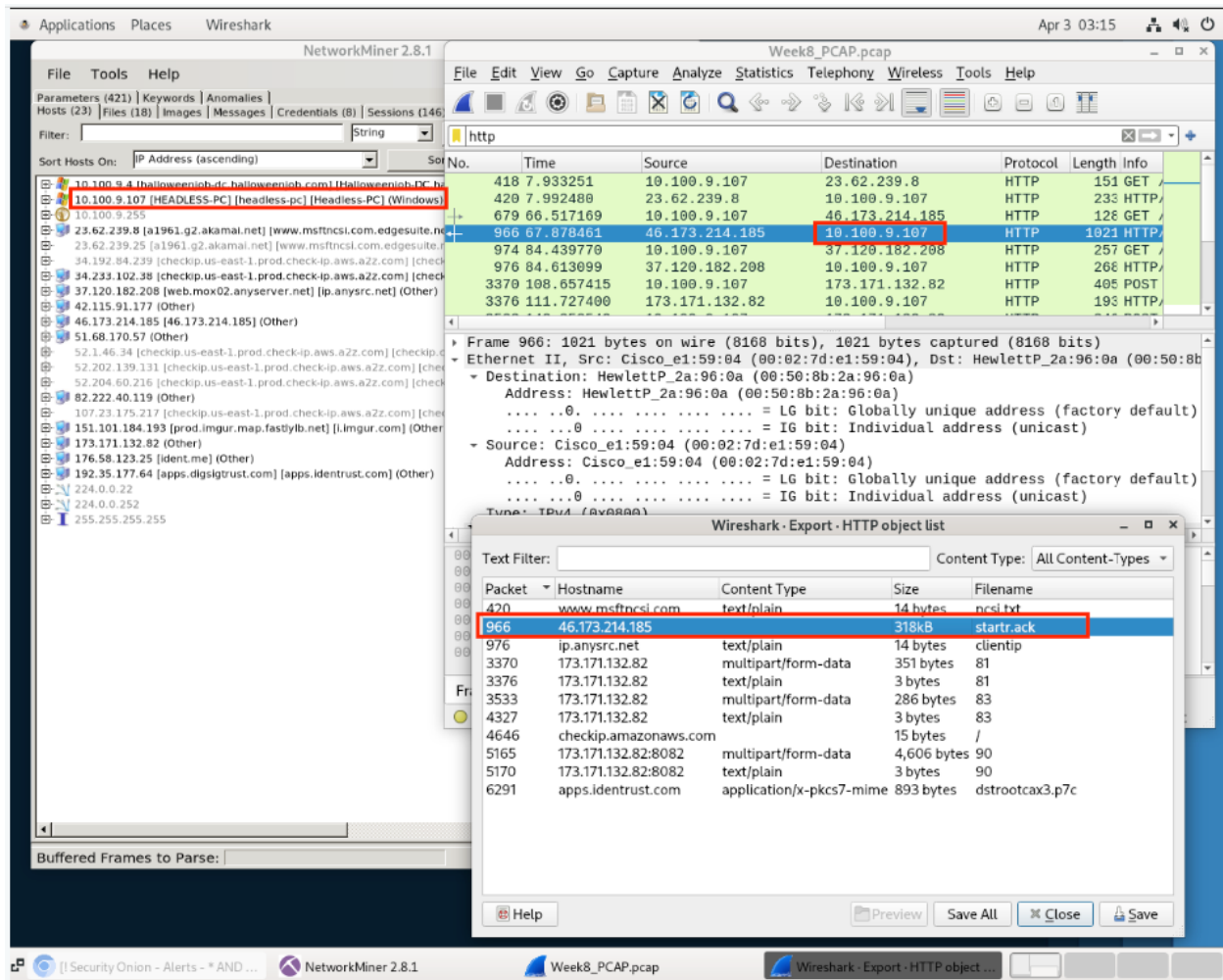
NetworkMiner identifies the suspicious transferred file as start.exe. The file details window displays the MZ header and PE-related structure, which is strong evidence that the transferred object is a Windows executable rather than ordinary document content.

Figure 2. VirusTotal result for the extracted file.



VirusTotal flagged the file as malicious with a 63/72 detection result in the screenshot. This does not replace packet analysis, but it strongly supports the conclusion that the downloaded executable is malicious.

Figure 3. Wireshark evidence of HTTP object transfer and follow-on POST traffic.



Wireshark shows the file transfer and multiple HTTP POST requests involving the suspicious communication path. POST traffic after the download is consistent with malware beaconing, command-and-control check-ins, or submission of host data. Based on the screenshot alone, it is safer to describe this as traffic consistent with possible exfiltration rather than proven full exfiltration.

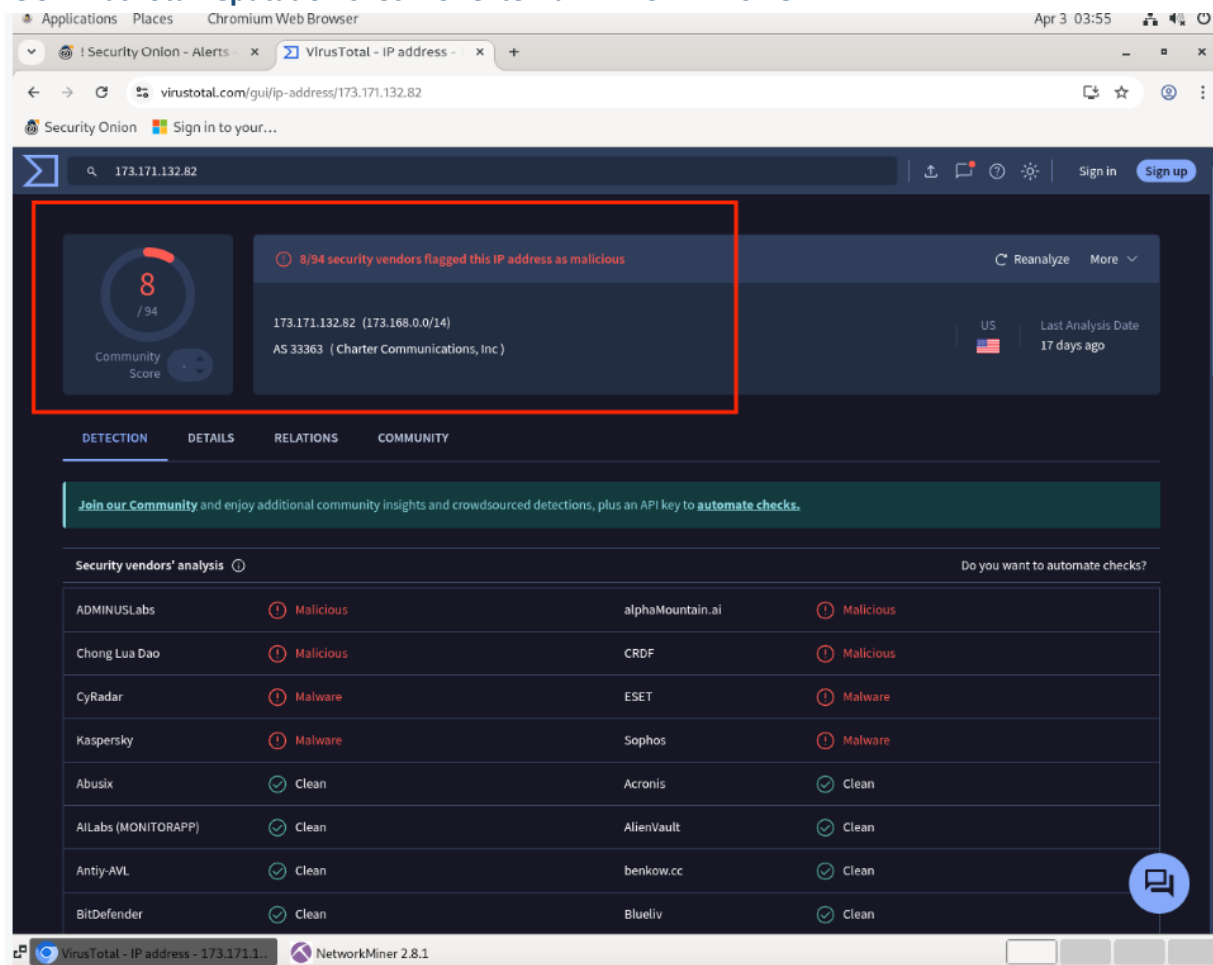
Figure 4. Security Onion alert tied to the suspicious connection.

The screenshot displays the Security Onion web interface in a Chromium Web Browser. The browser's address bar shows a URL starting with 'https://192.168.1.1/#/alerts?'. The interface has a dark theme and a sidebar on the left with navigation options like Overview, Alerts, Dashboards, Hunt, Cases, Detections, PCAP, Grid, Downloads, and Administration. The main content area is divided into two panels. The left panel shows a list of alerts, with the selected alert having a timestamp of '2018-10-31 15:36:17 UTC'. The right panel provides a detailed view of the selected alert, titled 'ET DROP Spamhaus DROP'. This panel includes a 'Summary' section with the text 'misc-attack', a 'Status' section showing 'Status: Enabled', and a 'Tuning' section with a '+', 'Enabled', 'Type', and 'Track/Regex' fields. Below these sections, a table lists various fields and their values, including 'message', 'network_community_id', 'network_data.decoded', 'network_packet_source', 'network_private_ip', 'network_public_ip', 'network_transport', 'observer name', 'related ip', 'rule.action', 'rule.category', 'rule.gid', 'rule.metadata.affected_product', 'rule.metadata.attack_target', and 'rule.metadata.confidence'. The bottom of the interface shows the version '2.4.150' and the license 'License: ELV2'.

Field	Value
message	["timestamp":"2018-10-31 15:36:17 UTC", "flow_id": "783578847031495", "pcap_cnf": "5185", "event_type": "ET DROP Spamhaus DROP", "signature": "ET MALWARE Trickbot Checkin Response", "category": "Malware C", "tag": "Major", "Description": "Generated By: Proofpoint_Nexus", "updated_at": "2024_05_04", "rule": "1", "pattern": "nocase: file data; content: '1/1/1'; startswith: ends with: reference:md5.5d2d5d96c6b1dc1d10bdcac", "Description": "Generated By: Proofpoint_Nexus, updated_at: 2024_05_04", "files": [{"filename": "sa91/HEAD", "Content-Type": "text/plain/vvvv1/1", "stream": "1", "packet": "AFCLKpYKAAJ94VKECABFAAAoNsAAIAGVC"}]
network_community_id	1rq+nVHtoAdBkVhEOqfn6H3aaww
network_data.decoded	HTTP/1.1 200 OK server: Cowboy date: Wed, 31 Oct 2018 15:36:10 GMT content-length: 3 Content-Type: text/plain
network_packet_source	/1/
network_private_ip	["10.100.9.107"]
network_public_ip	["173.171.132.82"]
network_transport	TCP
observer name	analystworkstation
related ip	["173.171.132.82", "10.100.9.107"]
rule.action	allowed
rule.category	Malware Command and Control Activity Detected
rule.gid	1
rule.metadata.affected_product	["Windows_XP_Vista_7_8_10_Server_32_64_Bit"]
rule.metadata.attack_target	["Client_Endpoint"]
rule.metadata.confidence	"High"

Security Onion records a timestamp of 2018-10-31 15:36:17 UTC and correlates internal host 10.100.9.107 with external IP 173.171.132.82. The alert is categorized as Malware Command and Control Activity Detected and shows high confidence, which independently corroborates the packet findings.

Figure 5. VirusTotal reputation check for external IP 173.171.132.82.



The related external IP was flagged by 8/94 security vendors in the screenshot. Reputation alone is not conclusive proof, but it adds context showing that the external system was already viewed as suspicious by multiple vendors.

Interpretation of the Malware Activity

The strongest evidence in this case is the combination of packet evidence, extracted artifact evidence, and independent alert/reputation evidence. The initial stage appears to be the delivery of a Windows executable to the internal endpoint. The next stage is visible as HTTP POST traffic from the affected host toward the suspicious external system. Security Onion then labels the activity as malware command-and-control related. Taken together, these artifacts support the conclusion that the endpoint was infected and communicated with an outside system after the download.

Secondary Infection Assessment

The follow-on behavior is consistent with a TrickBot-style secondary stage. TrickBot is widely documented as a modular trojan that can gather system information, steal credentials, maintain command-and-control communications, and download additional payloads (CISA, 2021; Microsoft, n.d.). In this capture, the observed HTTP POST traffic and the command-and-control alert fit that general pattern. Because the screenshots do not expose the full decrypted contents of every transfer, the most defensible conclusion is that the traffic is consistent with secondary malware activity rather than a definitive claim about every capability being used in this exact session.

Non-Technical Summary

In simple terms, a computer inside the network appears to have downloaded a malicious program from an outside server. After that, the same computer started talking back to the outside system in a way that looks like malware control traffic. The security tools used in this analysis all pointed in the same direction, so this was likely a real compromise and not normal web browsing.

References

- Cybersecurity and Infrastructure Security Agency. (2021). TrickBot Malware. CISA Cybersecurity Advisory AA21-076A. <https://www.cisa.gov/news-events/cybersecurity-advisories/aa21-076a>
- Microsoft. (n.d.). Win32/Trickbot threat description. Microsoft Security Intelligence. <https://www.microsoft.com/en-us/wdsi/threats/malware-encyclopedia-description?Name=Win32%2FTrickbot>
- VirusTotal screenshots and analyst observations captured during this investigation.